

Analytical statistical model for restaurant demand, revenue, and expenses

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Revised theory – consolidation draft

Abstract

This document proposes a parametric model for restaurant demand, revenue, expenses, and profit. The purpose is not to replace data analysis or machine learning, but to provide a transparent analytical reference framework whose assumptions are explicit and whose parameters can later be calibrated against observations. The presentation deliberately starts from the most general formulation, then introduces practical simplifications that keep the model analytically manageable, and finally ends with worked examples. The resulting theory is meant to be strong enough to guide later quantitative developments.

1 Introduction

The goal of these notes is to provide an analytical model for the expected financial behavior of a restaurant, starting from a description of the accessible population, the temporal structure of demand, the menu, and the expense model.

The present document does not use machine learning for prediction. Instead, it describes a structured parametric model whose ingredients can later be estimated, challenged, or refined from data. This is useful for early-stage exploration because it exposes the points where the business is most sensitive: catchment size, competition, service timing, basket composition, and recurring costs. If proven sufficient, this can also serve as a base for training ML predictive models such that one may get forecasts in near real time.

The document follows three recurring levels. A general form, with weak assumptions and broad interpretability. Follows a practical reference model, chosen so that the model remains analytically and numerically manageable. Then we present worked examples, used as a pedagogical case first and as a richer near-implementation case second.

The model aims at expected values over a planning horizon. It is therefore suited to scenario analysis, parameter sensitivity, and the construction of reproducible examples. It is not a stochastic queueing model, and it does not attempt to predict short-term shocks, weather anomalies, or operational disruptions unless such effects are introduced explicitly as extra factors.

The reference theory intentionally remains compact. It ignores many second-order effects, such as social-network feedback, inventory constraints, dynamic pricing, endogenous waiting times, and stockout behavior. These are natural future extensions rather than part of the reference model.

2 Theory: from general form to practical model

2.1 Notation, units, and modeling conventions

Table 1 defines the main symbols used throughout the document. One important convention is fixed immediately: $N_c(t)$ denotes an expected customer arrival rate, with units of persons per unit time, rather than an instantaneous headcount. This choice is necessary because revenue and expense aggregates are obtained by integrating over time.

Table 1: Core notation used in the revised theory.

Symbol	Units	Meaning
$N_g(t)$	persons	Accessible population in group g , with g in $\{l, w, tr\}$ for local residents, workers/students, and transit.
$N_{tot}(t)$	persons	Total accessible population reservoir, equal to the sum of the group populations.
$N_c(t)$	persons / hour	Expected customer arrival rate. In a time bin Δt , the expected number of clients is approximately $N_c(t)\Delta t$.
$A_g(t)$	dimensionless	Attractiveness / conversion factor for group g . In the general theory it is factorized into interpretable components.
$I_g(r, \theta)$	dimensionless	Spatial influence kernel describing the fraction of a population at distance r and direction θ that can reasonably reach the restaurant.
$R_{\text{eff},g}$	km	Effective radius associated with the spatial influence kernel of group g .
$S_{\text{eff},g}$	km ²	Effective area associated with the chosen influence kernel.
$D_g(\tau_d)$	1 / hour	Daily arrival kernel for group g . It carries the rate interpretation in the reference model.
$W_g(\tau_w), Y_g(\tau_y)$	dimensionless	Weekly and yearly envelopes for group g . They modulate the daily kernel.
$\Pi(\mathbf{n} t)$	PMF	Probability mass function of customer baskets $\mathbf{n} = (n_L)_L$ conditional on an arrival at time t .
$\bar{p}(t)$	currency units / customer	Expected spend per arriving customer at time t .
$\bar{c}(t)$	currency units / customer	Expected production cost per arriving customer at time t .
$\mathcal{R}(\Delta t)$	currency	Total expected revenue over a horizon Δt .
$E_{\text{tot}}(\Delta t)$	currency	Total expected expenses over the same horizon.
$\mathcal{I}(\Delta t)$	currency	Expected income / profit, defined as revenue minus total expenses.
$M_g(t)$	h ⁻¹	General temporal modulation used before the separable reference form is introduced.
p_L, c_L	currency units	Unit price and unit production cost of menu item L .
n_r	count	Number of competitor restaurants in the equal-quality competition limit.
$\alpha_m(t), f_m$	dimensionless	Payment-method probability and fee fraction for method m .
ϕ_i	dimensionless	Commute-overlap fraction used in the practical de-duplication term for work site i .

Table 2: Compact summary of the main assumptions.

Choice	main assumption
Attractiveness	Factorized into interpretable terms, then reduced to time-constant effective factors in the reference model.
Spatial reach	Gaussian radial kernel with peak normalization and effective area $S_{\text{eff}} = 2\pi R_{\text{eff}}^2$.
Temporal structure	Separable daily, weekly, and yearly modulation. D carries units; W and Y are mean-one envelopes.
Purchase model	Multi-item basket model in general, reduced to practical item-level probabilities when needed.
Expenses	Production, fees, recurring, staff, and fixed or one-time costs all remain explicit in the general model.
Study cases	Pedagogical example first, richer example second, with transit kept as an explicit limit-case sensitivity.

2.2 General population reservoir and customer flow

Let $\mathcal{G} = \{l, w, tr\}$ denote the set of population groups: local residents, workers or students, and transit populations. The most general decomposition of the accessible population is

$$N_{tot}(t) = \sum_{g \in \mathcal{G}} N_g(t). \quad (1)$$

The total population reservoir is not yet the customer flow. A second layer is needed to describe how the accessible population converts into arrivals. We therefore introduce a group-dependent attractiveness factor $A_g(t)$, and a general temporal modulation $M_g(t)$ with units of inverse time. The expected customer arrival rate is then

$$N_c(t) = \sum_{g \in \mathcal{G}} N_g(t) A_g(t) M_g(t). \quad (2)$$

This distinction is essential. The quantities $N_g(t)$ and $N_{tot}(t)$ are counts of potentially reachable people, whereas $N_c(t)$ is a rate of arrivals. The model is easier to interpret when this distinction is made explicit from the outset.

Factorized attractiveness. In the general theory, attractiveness should not be treated as a single opaque scalar. It is more informative to write it as a product of interpretable factors,

$$A_g(t) = \prod_{m=1}^{M_g} A_g^{(m)}(t), \quad 0 \leq A_g^{(m)}(t) \leq 1. \quad (3)$$

Typical components may encode competition, visibility, offer fit, affordability, reputation, and convenience. The main simplification is to assume approximate orthogonality between these factors so that the product remains compact. In pedagogical examples or first implementations, this factorized expression can be reduced to a single effective factor A_g or even to a single shared A .

Competition limit. If the focal restaurant competes with n_r comparable restaurants inside the same catchment area, and if n_r counts competitors only, then a conservative equal-quality approximation is

$$A_g^{(\text{comp})} \simeq \frac{1}{n_r + 1}. \quad (4)$$

This relation is not a law; it is a limiting-case approximation obtained when the restaurants are almost interchangeable and the available customer demand is divided roughly evenly among them.

Reference model. For the reference model, we assume that the factorized attractiveness terms vary slowly enough to be treated as time-constant effective parameters over the planning horizon. Time-varying preference shocks can still be absorbed into the more general modulation $M_g(t)$ if needed. In the reference model, the temporal modulation is further factorized into daily, weekly, and yearly components,

$$N_c(t) = \sum_{g \in \mathcal{G}} N_g(t) A_g D_g(\tau_d) W_g(\tau_w) Y_g(\tau_y), \quad (5)$$

where $\tau_d = t \bmod 24 \text{ h}$, $\tau_w = t \bmod 7 \text{ d}$, and $\tau_y = t \bmod 1 \text{ y}$. When parameter count must be reduced, one may collapse this to a shared structure,

$$N_c(t) = A N_{tot}(t) D(\tau_d) W(\tau_w) Y(\tau_y). \quad (6)$$

This compact form is convenient, but the document keeps the group-dependent version as the conceptual reference.

2.3 Spatial influence and effective area

The number of residents, workers, or transit users that belong to the accessible reservoir depends on space. The most general object is an anisotropic influence kernel $I_g(r, \theta)$, which depends on radial distance, direction, terrain, road structure, and effective mobility.

For the living population, the general expression is

$$N_l = \int_0^{2\pi} \int_0^\infty \rho_l(r, \theta) I_l(r, \theta) r dr d\theta, \quad (7)$$

where $\rho_l(r, \theta)$ is the local residential density. In the reference model we now assume a uniform local density ρ_l and an isotropic kernel $I_l(r, \theta) = I_l(r)$.

Gaussian reference kernel. The reference kernel is chosen to be Gaussian and peak-normalized,

$$I_l(r) = \exp\left(-\frac{r^2}{2R_{\text{eff},l}^2}\right), \quad I_l(0) = 1. \quad (8)$$

It is important not to confuse peak normalization with integral normalization. Equation 8 is not a probability density over space; it is a dimensionless reach kernel whose peak value is one.

The corresponding effective area is

$$S_{\text{eff},l} = \int_0^{2\pi} \int_0^\infty I_l(r) r dr d\theta = 2\pi R_{\text{eff},l}^2. \quad (9)$$

Therefore the living population reservoir becomes

$$N_l = \rho_l S_{\text{eff},l} = 2\pi R_{\text{eff},l}^2 \rho_l. \quad (10)$$

In the reference model, local residents are treated as a time-invariant reservoir to first order, so $N_l(t) \approx N_l$ over the planning horizon unless a separate residential time modulation is introduced. If a probability-normalized kernel is preferred, one may instead use $I_l/S_{\text{eff},l}$, but then the interpretation of the integral changes and an external area scale must be reintroduced elsewhere. For the present document, peak normalization is retained for the spatial kernel.

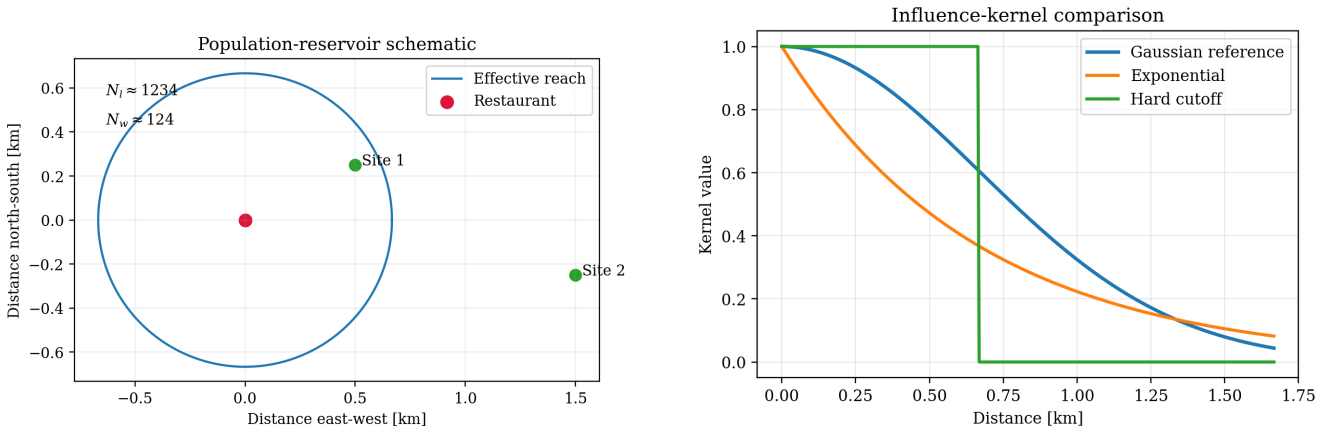


Figure 1: Left: conceptual schematic of the population reservoir, the restaurant, nearby work sites, and the effective spatial reach. Right: comparison of three radial kernels. The Gaussian kernel is retained in the reference formulation because it is smooth and analytically tractable.

Working population. A working or studying site is better modeled as a discrete reservoir. Let $N_{w,i}(t)$ be the total number of workers or students at site i , and let $N_{w,i}^{(0)}(t)$ denote the subset already counted among the local residents. The practical reference form is

$$N_w(t) = \sum_{i=1}^{n_w} \left(N_{w,i}(t) - N_{w,i}^{(0)}(t) \right) I_w(r_i). \quad (11)$$

This is a compact de-duplication rule: residents who also work nearby are not double-counted when constructing the accessible reservoir. In the practical study cases, we write $N_{w,i}^{(0)}(t) = \phi_i N_{w,i}(t)$, where $\phi_i \in [0, 1]$ is the commute-overlap fraction of site i . More detailed versions could replace ϕ_i by an overlap integral between residential density and work-site catchment structure.

Transit population. Transit demand deserves a separate term in the general theory. It may arise from tourists, transport hubs, beaches, or other non-resident flows. In a full model one would infer $N_{tr}(t)$ from local pedestrian flux or transport exposure. In the reference study case below, the clean limit $N_{tr}(t) = 0$ is kept first, and a small transit extension is discussed afterward.

Alternative kernels. The Gaussian reference kernel is not the only possible choice. An exponential kernel has a longer tail and may better reflect corridors or sparse catchments; a hard cutoff is easy to explain but introduces a non-smooth boundary. Figure 1 is therefore not merely decorative: it clarifies why the Gaussian choice is retained in the reference formulation while keeping alternatives visible for later refinements.

2.4 Daily, weekly, and yearly modulation

Demand varies across several recurring timescales. The most general object is a non-separable modulation $M_g(t)$ in equation 2. This is conceptually clean but often too rich for analytical work. The reference approximation is therefore the separable form of equation 5.

Daily kernel. The daily component carries the rate interpretation in the reference model. We define it so that its integral over one day yields an expected number of visits per accessible person,

$$\int_0^{24 \text{ h}} D_g(\tau_d) d\tau_d = \nu_g, \quad (12)$$

where ν_g is the expected number of daily meal opportunities or visits for group g . A practical parametric form is a sum of Gaussian meal peaks,

$$D_g(\tau_d) = \frac{\nu_g}{Z_g} \sum_{m=1}^{M_g^{(\text{meal})}} a_{g,m} \exp \left(-\frac{(\tau_d - \mu_{g,m})^2}{2\sigma_{g,m}^2} \right), \quad (13)$$

with normalization constant

$$Z_g = \int_0^{24 \text{ h}} \sum_{m=1}^{M_g^{(\text{meal})}} a_{g,m} \exp \left(-\frac{(\tau_d - \mu_{g,m})^2}{2\sigma_{g,m}^2} \right) d\tau_d. \quad (14)$$

This choice preserves a simple analytical description of repeated meal peaks while keeping the daily structure explicit.

Weekly and yearly envelopes. The weekly and yearly components should modulate the daily kernel without changing its physical units. A coherent reference convention is therefore

$$\frac{1}{T_w} \int_0^{T_w} W_g(\tau_w) d\tau_w = 1, \quad T_w = 7 \text{ d}, \quad (15)$$

and

$$\frac{1}{T_y} \int_0^{T_y} Y_g(\tau_y) d\tau_y = 1, \quad T_y = 365 \text{ d}. \quad (16)$$

With this convention, W_g and Y_g are dimensionless envelopes and D_g retains the full rate interpretation.

A practical weekly envelope can be written as day-of-week weights renormalized to unit mean,

$$W_g(\tau_w) = \frac{\omega_{g,d(\tau_w)}}{\frac{1}{7} \sum_{k=1}^7 \omega_{g,k}}, \quad (17)$$

where $d(\tau_w)$ is the day-of-week index. Such a weekly modulation can be constructed by repeating the daily template and weighting the days of the week appropriately.

A simple yearly envelope is

$$Y_g(\tau_y) = A_{y,g} - D_{y,g} \cos \left(\frac{2\pi\tau_y}{P_g} + \phi_g \right), \quad A_{y,g} > D_{y,g} \geq 0. \quad (18)$$

This is intentionally simple. Holidays, events, or weather shocks can later be added as local pulses or exogenous corrections on top of this yearly envelope.

Table 3: Normalization choices for the temporal modulators.

Choice	Definition	Comment
Full integral normalization	Each of D, W, Y integrates to one over its own period.	Mathematically elegant, but the product DWY then needs an extra intensity scale to recover a rate.
Mixed normalization (reference)	D integrates to ν ; W and Y have mean one over their period.	Best compromise for theory plus implementation: D carries the units, W and Y remain interpretable envelopes.
Peak normalization	Each component is normalized so that its maximum equals one.	Useful for plotting and rough comparisons, but poor for integrated quantities.

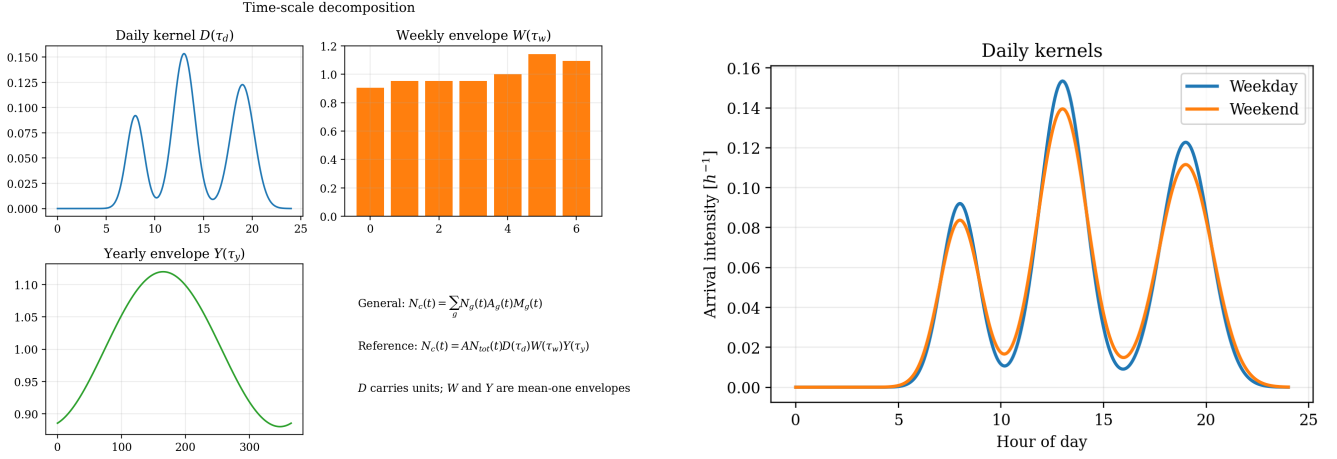


Figure 2: Left: schematic of the general-to-practical decomposition from a population reservoir to a time-dependent arrival rate. Right: example daily kernels for a weekday and a weekend day.

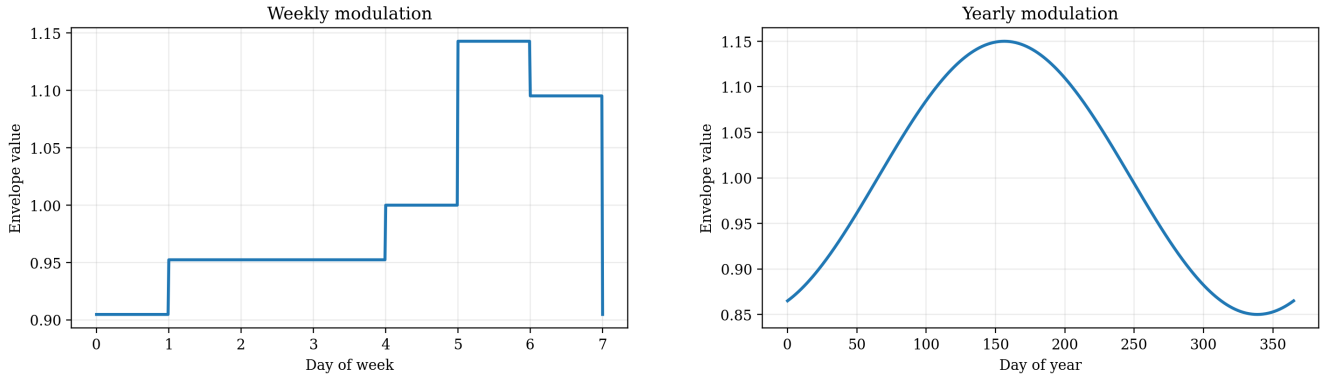


Figure 3: Example weekly and yearly envelopes used in the worked examples. In the reference convention the daily kernel carries the units, while the weekly and yearly curves have unit mean over their own period.

3 Menu, baskets, and expected spend

The menu is not only a list of prices. It is a time-dependent random object because basket composition changes across services and across customer groups. The correct starting point is therefore the basket distribution itself.

3.1 General basket model

Let L index menu items and let $\mathbf{n} = (n_L)_L$ denote the random basket associated with one arriving customer at time t . The most general object is the conditional basket distribution $\Pi(\mathbf{n}|t)$, with $n_L \in \mathbb{N}$. The expected spend per customer at time t is then

$$\bar{p}(t) = \sum_{\mathbf{n}} \Pi(\mathbf{n}|t) \sum_L n_L p_L, \quad (19)$$

and the expected production cost per customer is

$$\bar{c}(t) = \sum_{\mathbf{n}} \Pi(\mathbf{n}|t) \sum_L n_L c_L. \quad (20)$$

This formulation keeps multi-item purchases in the theory from the outset. It is therefore the appropriate reference formulation for the present document.

3.2 Practical approximation used for analysis

The full basket distribution is usually too large to parameterize directly. A practical reduction is to work with per-item expected multiplicities or, under a stronger assumption, with at-most-one-unit Bernoulli probabilities. Let $\pi_L(t)$ denote the probability that one unit of item L is included in the basket at time t . Under the at-most-one-unit approximation,

$$\bar{p}(t) = \sum_L p_L \pi_L(t), \quad \bar{c}(t) = \sum_L c_L \pi_L(t). \quad (21)$$

This reduced description preserves a tractable connection between service-dependent choice probabilities and expected monetary flows.

A service-specific parameterization can be written as

$$\pi_L(t) = q_L s_{\sigma(L)}(\tau_d), \quad (22)$$

where q_L is an item-level inclusion probability over the day, $\sigma(L)$ identifies the relevant service, and s_σ is a normalized service-time profile satisfying

$$\int_0^{24 \text{ h}} s_\sigma(\tau_d) d\tau_d = 1. \quad (23)$$

The general multi-item theory is therefore reduced to a manageable special case without losing interpretability.

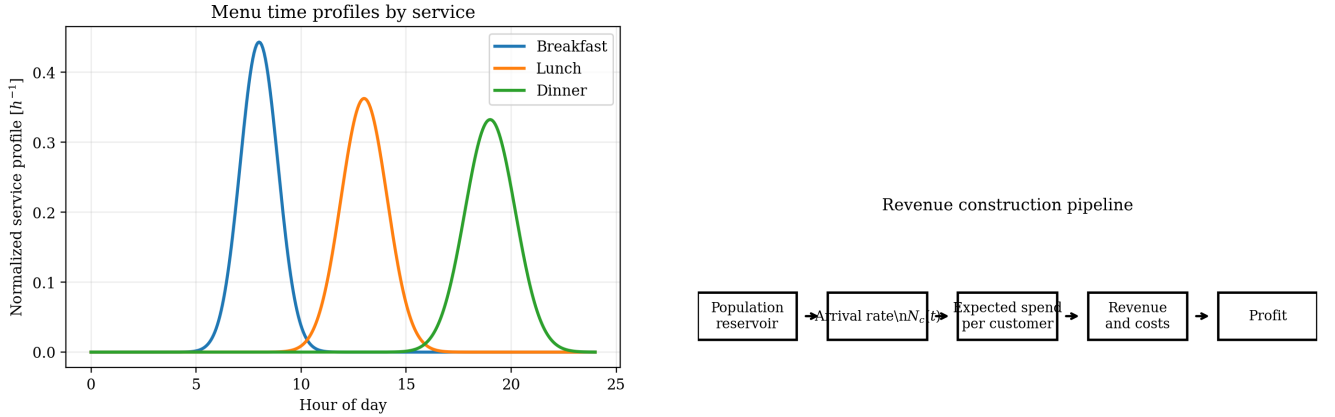


Figure 4: Left: illustrative service-specific menu time profiles. Right: conceptual pipeline from population reservoir to customer flow, expected spend, revenue, expenses, and profit.

4 Revenue, expenses, and profit

4.1 Expected revenue

With the notation above, the expected revenue rate is simply the customer arrival rate multiplied by the expected spend per customer,

$$r(t) = N_c(t) \bar{p}(t). \quad (24)$$

The expected revenue over a horizon Δt is therefore

$$\mathcal{R}(\Delta t) = \int_0^{\Delta t} N_c(t) \bar{p}(t) dt. \quad (25)$$

This is a proper equality, not a proportionality. The normalization choices are carried by D, W, Y and by the basket model itself; there is no missing multiplicative constant.

A discrete-time approximation with step δt is

$$\mathcal{R}(\Delta t) \approx \sum_{n=0}^{N-1} N_c(t_n) \bar{p}(t_n) \delta t, \quad (26)$$

where $t_n = n\delta t$ and $N\delta t = \Delta t$. This provides a natural discrete approximation for numerical evaluation.

4.2 Production cost and payment fees

The expected production-cost rate is

$$e_{\text{prod}}(t) = N_c(t) \bar{c}(t), \quad (27)$$

so that

$$E_{\text{prod}}(\Delta t) = \int_0^{\Delta t} N_c(t) \bar{c}(t) dt. \quad (28)$$

If several payment methods are available, with method probabilities $\alpha_m(t)$ and fee fractions f_m , the expected fee factor is

$$\bar{f}(t) = \sum_m \alpha_m(t) f_m, \quad (29)$$

and the corresponding fee expense is

$$E_{\text{fees}}(\Delta t) = \int_0^{\Delta t} N_c(t) \bar{p}(t) \bar{f}(t) dt. \quad (30)$$

The deterministic expectation of equation 30 is the natural analytical object. If desired, one may later replace it with a stochastic simulation of payment choices.

4.3 Recurring, staff, and fixed costs

Recurring costs are most naturally expressed as rates. Let r_i be the recurring expense rate associated with item i (rent, software, utilities, marketing, accounting, and so on). Then

$$E_{\text{rec}}(\Delta t) = \sum_i r_i \Delta t. \quad (31)$$

Staff costs can be written either from hourly schedules or from average staffing rates. If staff member j is paid w_j per hour and works $h_j(\Delta t)$ hours during the horizon,

$$E_{\text{staff}}(\Delta t) = \sum_j w_j h_j(\Delta t). \quad (32)$$

The most general model also includes fixed or one-time costs. If F_k denotes the one-time expense associated with event k , then

$$E_{\text{fix}}(\Delta t) = \sum_{k \in \mathcal{K}(\Delta t)} F_k, \quad (33)$$

where $\mathcal{K}(\Delta t)$ is the set of fixed-cost events occurring within the horizon. A reduced operating model may omit this term intentionally, but the general model should keep it explicit.

The total expense model is therefore

$$E_{\text{tot}}(\Delta t) = E_{\text{prod}}(\Delta t) + E_{\text{fees}}(\Delta t) + E_{\text{rec}}(\Delta t) + E_{\text{staff}}(\Delta t) + E_{\text{fix}}(\Delta t), \quad (34)$$

and expected income / profit is

$$\mathcal{I}(\Delta t) = \mathcal{R}(\Delta t) - E_{\text{tot}}(\Delta t). \quad (35)$$

5 Study cases

The figures used in this section are generated from external scripts and parameter files documented in Appendix B and cited explicitly in the bibliography. The purpose of the first example is pedagogical. The second one is richer and introduces a more detailed set of assumptions while preserving analytical readability. The optional transit sensitivity remains explicit: the first figures use $N_{tr} = 0$, and a small nonzero transit term, such as 5%, can be examined as a secondary scenario.

5.1 Pedagogical example

The pedagogical example uses a single shared customer group and no transit component. It is intended to clarify the meaning of the formulas before richer heterogeneity is introduced.

Table 4: Pedagogical example used to illustrate the reference equations.

Parameter	Value	Interpretation
N_{tot}	220 persons	Single shared accessible reservoir.
A	0.18	Product of competition, visibility, and offer-fit factors.
ν	1.0 day^{-1}	One expected daily visit opportunity per accessible person.
Daily peaks	8h, 13h, 19h	Breakfast, lunch, and dinner structure.
Weekly weights	[0.95, 1.00, 1.00, 1.00, 1.05, 1.20, 1.15]	Higher envelope from Friday to Sunday.
Yearly envelope	1.0 $0.12 \cos(2\pi\tau_y/365 \text{ d})$	Mild annual variation.
Transit term	0	Clean limit case.

In this first example, the menu is reduced to a few representative items with clear service assignments. The main purpose is to connect equations 6, 21, and 25 to directly readable plots.

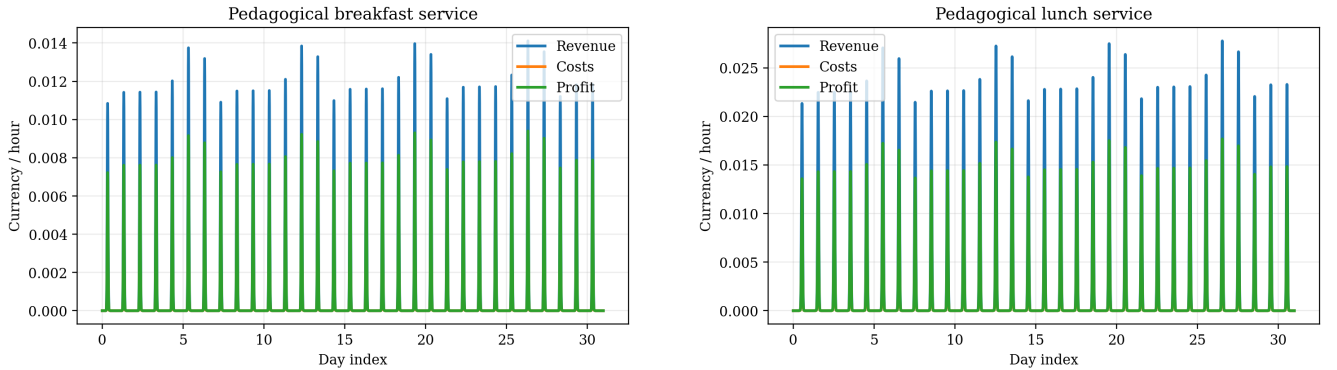


Figure 5: Pedagogical example: time-resolved revenue, cost, and profit contributions for two representative services. These plots are meant to be simple enough to translate later into property-style tests or narrow numerical checks.

5.2 Richer example

The second example keeps the same structure but introduces a richer set of scales and parameters. In particular, it uses a Gaussian spatial kernel, a radius inferred from travel time and travel speed, a competition factor based on n_r , and a simple harmonic yearly envelope.

Table 5: Richer example with parameters chosen to remain close to the richer analytical setting considered here.

Parameter	Value	Interpretation
Residential density ρ_t	442 persons / km ²	Same order of magnitude as the current setup file.
Effective travel time radius	10 min	Baseline mobility horizon.
Travel speed	4 km / h	Gives $R_{\text{eff}} \approx 0.67$ km.
Competition count n_r	20	Gives $A^{(\text{comp})} \approx 1/21$.
Working sites	2	One nearby work site and one farther education site.
Commute overlap fraction	15%	Removes the part already counted in residents.
Transit term	0, then 5% extension	Limit case first, small extension second.
Daily visit scale ν	1.0 day ⁻¹	Baseline average.
Yearly envelope	simple cosine family	Matches the chosen harmonic family before adding holiday pulses.

Using equations 10 and 11, this richer example remains analytically readable while already matching the spatial structure more clearly. It therefore provides a natural richer counterpart to the pedagogical case.

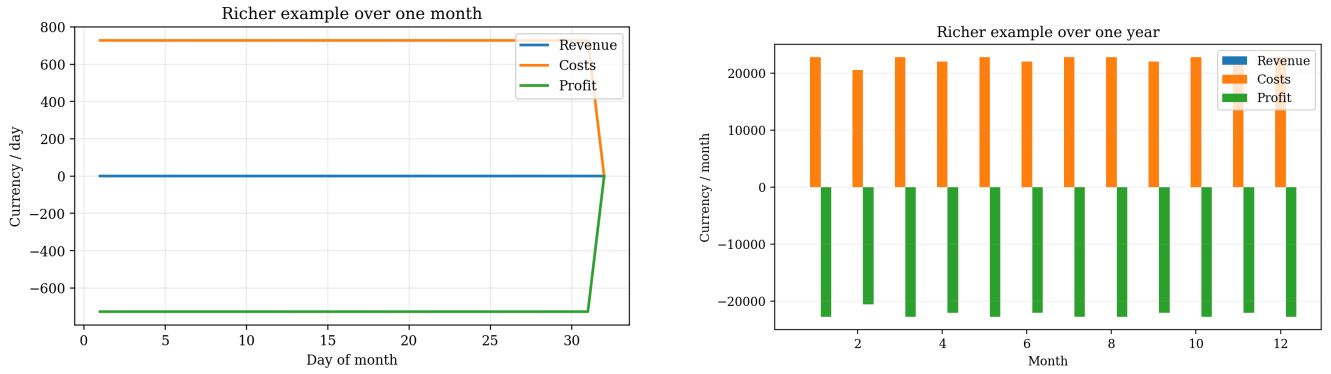


Figure 6: Richer example: total revenue, total costs, and profit over a monthly window and over an annual summary. The optional 5% transit extension can be explored as a sensitivity layer without changing the general structure of the model.

6 Discussion and practical limits

The revised model is intentionally layered. The general form keeps the conceptual objects clear: population reservoir, conversion structure, temporal modulation, basket model, and expense decomposition. The reference model then chooses a Gaussian spatial kernel, a separable daily-weekly-yearly structure, and a compact basket approximation because these assumptions remain analytically tractable while preserving the main mechanisms of interest.

Several limitations remain explicit. The separable temporal structure remains an approximation, and holidays, events, or weather shocks may introduce couplings between timescales that are absent from the reference formulation. The factorization of attractiveness likewise assumes an approximate orthogonality between several interpretable effects; this may cease to be adequate when competition, accessibility, and customer preferences become strongly dependent on one another. The practical menu approximation also ignores within-basket correlations unless the full basket distribution is retained. Finally, transit demand remains strongly location-dependent, and the reduced operating view may omit one-time costs when attention is restricted to short-run operation, even though the general expense equation keeps them explicitly.

These limitations are not flaws of presentation. They indicate the directions in which the theoretical framework can be refined once additional data, constraints, or modeling priorities are introduced.

A Appendix A: analytical summary for numerical realization

Although the present article is purely theoretical, it is useful to isolate the analytical blocks that any later numerical realization would have to reproduce. The first block concerns the spatial sector, namely the evaluation of the Gaussian influence kernel, its associated effective area, and the construction of the living and working reservoirs. The second block concerns temporal modulation, that is, the normalization of the daily meal-peak superposition together with the weekly and yearly envelopes. The third block concerns the basket description, where the general multi-item formulation may be reduced to expected item-level contributions when a simpler approximation is desired. The fourth block concerns the aggregation stage, in which arrival rates, expected spend, and expense rates are integrated over the chosen horizon. In that sense, the theory naturally decomposes into a spatial block, a temporal block, a basket block, and an aggregation block, each of which can later be translated into a concrete numerical procedure without modifying the equations themselves.

B Appendix B: figure reproducibility

All figures referenced in this article are meant to be reproducible. The external script responsible for figure generation and the associated parameter file are treated here as bibliographic objects and cited accordingly [1, 2]. In practice, reproducibility requires three ingredients: a parameter file defining the pedagogical and richer examples, a script that generates the conceptual figures and numerical plots, and a stable output location in which the generated graphics are written before compilation. If one wishes to examine the transit sensitivity separately, the same external material can be reused with a modified transit contribution. This bibliographic traceability preserves the scientific status of the document itself while keeping the numerical material externally identifiable and reproducible.

References

- [1] Othman Benomar. Figure-generation script for the restaurant theory document, 2026. Repository script at `docs/tex/scripts/generate_theory_figures.py`.
- [2] Othman Benomar. Parameter file for the restaurant theory figures, 2026. Repository data file at `docs/tex/scripts/theory_figure_params.json`.